

Experiment No: **01**

Experiment Name: **Introduction to Basic Bash Commands and File System Management.**

Theory: Bash is a command-line interpreter that allows users to interact with the Linux operating system by executing text-based commands. It provides direct control over file system management through commands that navigate directories, create and modify files, and manage system resources. Commands like **`pwd`** and **`cd`** handle navigation, while **`ls`** lists contents. **`mkdir`** and **`touch`** create directories and files, **`nano`** edits them, and **`cat`** displays contents. **`rm`** and **`rmdir`** remove files and directories, enabling efficient system administration.

Command	Description
whoami	Displays the current username of the logged-in user.
pwd	Prints the present working directory (full path of current location).
ls	Lists all files and directories in the current directory.
cd	Changes the current directory to the home directory.
cd ..	Moves up one level to the parent directory.
clear	Clears the terminal screen of all previous text and output.
mkdir	Creates a new empty directory.
mkdir folder1 folder2	Creates multiple directories at once.
touch filename.extension	Creates a new empty file with the specified name.
cat filename.extension	Displays the contents of a file on the terminal.
nano	Opens the nano text editor for creating or editing files.
rm	Removes (deletes) a file.
rmdir	Removes an empty directory.
rm -rf	Forcefully removes files and directories recursively (use with caution).

Commands and Output:

➤ whoami

```
> whoami
atif
```

➤ `cd`

```
> cd Documents/study
```

```
~/Documents/study
```

➤ pwd

```
> pwd
/home/atif/Documents/study
```

```
└─┬─┐ ~/Documents/study ───
└─┬─┐
```

➤ cd ..

```
/home/atif/Documents/study
> cd ..
```

```
└─┬─┐ ~/Documents ───
└─┬─┐
```

➤ cd Tab

```
/
```

```
└─┬─┐ ~/Documents ───
└─┬─┐ > cd study/
```

➤ mkdir

```
> mkdir testife
```

```
└─┬─┐ ~/Documents/study ─
└─┬─┐
```

➤ ls

```
ls
-rwxr-xr-x - atif 26 Jul 22:11 others
-rwxr-xr-x - atif 26 Jul 22:06 testife
```

```
- └─┬─┐ ~/Documents/study ─
-> └─┬─┐
```

➤ mkdir foldernames

```
> cd testife
> mkdir testfolder1 testfolder2 testfolder3
> ls
drwxr-xr-x - atif 26 Jul 22:16 testfolder1
drwxr-xr-x - atif 26 Jul 22:16 testfolder2
drwxr-xr-x - atif 26 Jul 22:16 testfolder3
```

```
└─┬─┐ ~/Documents/study/testife ─
└─┬─┐
```

➤ Before

```
> exa -ls
error: a value is required for '--sort <FIELD>' but none was supplied
[possible values: name, Name, .name, .Name, size, ext, Ext, date, age, changed, accessed,
created, inode, type, none]
```

For more information, try '--help'.

```
> exa -l
.rw-r--r-- 1.8k atif 28 Jun 19:48 241311059_B.sql
.rw-r--r-- 44M atif 23 Jun 19:46 'ELECTRONIC COMMUNICATION SYSTEM BY GEORGE KENNEDY 2.pdf'
drwxr-xr-x - atif 21 Jul 08:17 'Questions 6th semester'
.rw-r--r-- 393 atif 25 Jul 15:29 script.js
drwxr-xr-x - atif 26 Jul 22:06 testife
>
>
> ls
drwxr-xr-x - atif 26 Jul 22:11 others
drwxr-xr-x - atif 26 Jul 22:06 testife
```

```
└─┬─┐ ~/Documents/study ───
└─┬─┐
```

➤ clear

```
└─┬─┐ ~/Documents/study ─── global 22:15:51
└─┬─┐
```

> rmdir

```
> rmdir testfolder1 testfolder2 testfolder3  
> ls
```

```
└─ ^ ▸ ~/Documents/study/testife —  
└─ > █
```

▸ touch

```
> touch myInfo.txt
```

```
└─ ^ ▸ ~/Documents/study/testife —  
└─ > █
```

▸ nano

```
> nano myInfo.txt
```

```
└─ ^ ▸ ~/Documents/study/testife —  
└─ > █
```

▸ cat

```
> cat myInfo.txt  
Name: Abdullah Atif  
Id: 241311051  
Section: B  
Semester: 6th  
Batch: 34th  
VUCSE, 2026(Summer)
```

```
└─ ^ ▸ ~/Documents/study/testife —  
└─ > █
```

▸ rm

```
> rm myInfo.txt  
> ls
```

```
└─ ^ ▸ ~/Documents/study/testife —  
└─ > █
```

▸ rm -rf

```
> rm -rf testife
```

```
└─ ^ ▸ ~/Documents/study —  
└─ > █
```

Discussion:

This experiment demonstrated the role of Bash commands in Linux file system management. Commands like `pwd`, `cd`, and `ls` enabled directory navigation, while `mkdir` and `touch` facilitated file creation. `Nano` and `cat` allowed content viewing and editing, whereas `rm` and `rmdir` handled deletion. Using `rm -rf` highlighted the need for caution with recursive deletion to prevent data loss. We acquired practical skills in basic Bash operations for Linux administration. These commands provide a solid foundation for advanced scripting and command-line workflow.